Hotel Booking Demand Clustering Study

A Reproducible Experimental Study of Clustering

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Abstract

This report presents a reproducible clustering study of the Hotel Booking Demand dataset. The goal is to identify interpretable booking-level segments using only pre-stay observable features, while excluding outcome and post-event variables to prevent leakage. We compare K-Means, iK-Means, and Ward hierarchical clustering using internal validity indices, stability analysis, and preprocessing sensitivity checks. The selected solution is K-Means with $k = 3$ on the value-inclusive representation, producing three interpretable profiles: Non-Refundable Advance Booker, Standard Flexible City Guest, and Premium Resort Long-Stay Guest. The results suggest that deposit commitment, stay duration, and booking value provide useful axes for hotel revenue-management segmentation, although conclusions are limited by the absence of ground-truth labels and the booking-level unit of analysis.

Contents

1 Introduction

1.1 Context

Hotel revenue management depends on understanding who books, when they book, and how their bookings differ in value and service requirements. Booking records contain pre-arrival signals such as rate, stay duration, booking horizon, deposit commitment, meal plan, parking need, and special requests. These signals can reveal latent booking types that support pricing, upselling, and operational planning decisions.

1.2 Segmentation question

The concrete segmentation question addressed in this project is:

> Can hotel booking records be partitioned into distinct and interpretable behavioural segments using only pre-stay observable characteristics, such as daily rate, total estimated spend, length of stay, booking horizon, meal plan, deposit commitment, special requests, and parking need, in order to identify guest profiles that support targeted revenue-management and personalised-service strategies?

The unit of analysis is the individual booking record. A guest may appear more than once in the dataset, so the clusters should be interpreted as booking segments rather than permanent guest identities.

1.3 Why clustering is appropriate

No ground-truth segment labels are available. The aim is therefore discovery rather than prediction, making unsupervised clustering appropriate. Clustering is also suitable because the dataset contains heterogeneous booking behaviours in lead time, price, stay duration, deposit type, and service requests. To avoid leakage, only variables available at the time of booking are used as clustering inputs; outcome and post-event variables are reserved for post-hoc interpretation only.

1.4 Research questions

The report addresses the course research questions as follows:

- **RQ1 – Problem framing:** define a booking-level segmentation question and justify clustering.
- **RQ2 – Representation:** define leakage-safe preprocessing and feature representations for mixed-type booking data.
- **RQ3 – Modelling:** compare K-Means, iK-Means, and Ward hierarchical clustering.
- **RQ4 – Interpretation:** profile the discovered clusters as booking segments.
- **RQ5 – Robustness:** evaluate sensitivity to seeds, subsampling, scaling, feature weighting, and representation changes.

2 Dataset and Preprocessing

2.1 Dataset documentation

Table 1: Dataset documentation and traceability.

Item	Details
Dataset	Hotel Booking Demand
Version	Course release v1
Unit of analysis	One row corresponds to one booking record
Original size	119,390 rows and 32 columns
Time span	Arrivals from July 2015 to August 2017
Source	António, de Almeida, and Nunes (2019); dataset provided through the course platform
Raw-data governance	Raw CSV is not committed to the repository; it must be placed in `data/` according to the repository instructions
Checksum	SHA-256: `7c2ae42a...c7b1fc06`
Usage terms	Dataset used under the course release terms; no raw data redistributed in the repository

2.2 Data quality

The main data-quality issues identified are:

- `children` has 4 missing values, imputed as 0 because missing children counts are interpreted as no children in the booking.
- `country` has 488 missing values and is retained only for post-hoc profiling, not as a clustering input.
- `agent` and `company` have high missingness and high cardinality; both are dropped as ID-like fields.
- `adr` has a heavy right tail and one negative value, requiring robust preprocessing and careful interpretation.

- Rows with `adr = 0` are excluded from clustering because they correspond to complementary or test-like bookings. This removes 1,960 records, leaving 117,430 analytic records.

Table 2: Main missingness and governance decisions.

Variable	Missing	Role	Decision
`children`	4	Numerical	Impute as 0
`country`	488	Categorical	Exclude from clustering; retain for profiling
`agent`	16,340	ID-like categorical	Drop due to high missingness/cardinality
`company`	112,593	ID-like categorical	Drop due to high missingness/cardinality

2.3 Leakage control

The segmentation index time is the booking creation time. Therefore, only variables available at or before booking time can be used to form clusters. The following variables are excluded from clustering inputs:

- `is_canceled`: outcome variable.
- `reservation_status`: post-decision/post-arrival status.
- `reservation_status_date`: date associated with the realised status.
- `agent` and `company`: high-cardinality ID-like fields.
- `country`: excluded from clustering to avoid geographical profiling effects; retained only for post-hoc descriptive analysis.

2.4 Feature engineering

The final clustering features are:

- `total_nights` = `stays_in_week_nights` + `stays_in_weekend_nights`.
- `total_spend_log` = $\log(1 + \texttt{adr}) \times$ `total_nights`, used as a compressed estimated-spend proxy.
- `is_non_refund` = 1 if `deposit_type` is Non Refund, 0 otherwise.
- `lead_time_rank`, an ordinal quantile binning of `lead_time`: 0 for last-minute, 1 for mid-horizon, and 2 for advance bookings.
- `meal_encoded`, an ordinal encoding of meal-plan inclusivity: SC/Undefined = 0, BB = 1, HB = 2, FB = 3.
- `total_of_special_requests`.
- `required_car_parking_spaces`.

2.5 Representations and distance

Two Euclidean representations are studied:

Table 3: Final representations.

Representation	Definition
R-EUCLID-robust-weighted-withADR	**Features:** `total_spend_log`, `total_nights`, `is_non_refund`, `total_of_special_requests`, `required_car_parking_spaces`, `meal_encoded`, `lead_time_rank`. **Scaling/weights:** RobustScaler on continuous variables; `is_non_refund` $\times 3.0$; `meal_encoded` $\times 0.5$.
R-EUCLID-robust-weighted-noADR	**Features:** same as R-EUCLID-robust-weighted-withADR, excluding `total_spend_log`. **Scaling/weights:** same scaling and weights as R-EUCLID-robust-weighted-withADR.

All models and internal validity indices are computed in the same weighted Euclidean space used to fit the corresponding clustering model.

3 Clustering Methods

3.1 K-Means baseline

K-Means is used as the mandatory baseline. The predefined search interval is $k \in \{2, \ldots, 10\}$, chosen because $k = 2$ is the minimum meaningful segmentation and values above 10 are unlikely to remain managerially interpretable. The configuration is `n_init=10`, `max_iter=300`, and `random_state=42` for the main run.

3.2 iK-Means

iK-Means is included as an automatic cluster-discovery and initialisation-oriented method. In the current experiment log, iK-Means results must be treated carefully because the uploaded `experiments.csv` contains a skipped iK-Means row. If the final repository contains a successful iK-Means run, its metrics should replace the TODO values in Table **??** and be logged consistently in `experiments.csv`. If not, the report should explicitly state that iK-Means failed and explain the limitation.

3.3 Agglomerative hierarchical clustering

Agglomerative hierarchical clustering with Ward linkage is used as the alternative clustering family. Ward linkage is compatible with the chosen representation because the features are embedded in a Euclidean vector space. Due to computational cost, Ward clustering is run on a reproducible subsample of 30,000 records. The main subsample uses seed 42, and robustness is assessed using additional subsamples.

4 Experimental Protocol

4.1 Internal validity

Models are evaluated using:

- **Silhouette score** as the primary compactness/separation index.
- **Calinski–Harabasz index** as an additional compactness/separation index, where higher is better.
- **Davies–Bouldin index**, where lower is better.

4.2 Family-specific diagnostics

For hierarchical clustering, a Ward dendrogram is used to inspect merge structure and support the choice of the cut. For iK-Means, the automatic number of clusters and any anomalous extracted clusters should be documented.

4.3 Stability and sensitivity

The robustness protocol includes:

- K-Means repeated across 10 random seeds for $k \in \{2, 3, 4, 5\}$.
- Adjusted Rand Index (ARI) between runs as the stability measure.
- Hierarchical clustering repeated on controlled subsamples.
- RobustScaler versus StandardScaler as a preprocessing sensitivity check.
- withADR versus noADR comparison to test whether clusters are driven only by price.
- PCA representation study as Extension Module E4.

4.4 Reproducibility

All experiments are intended to be reproducible from the repository using:

- `experiments.csv` for experiment traceability.
- `environment.yml` or equivalent lockfile for dependencies.
- `run_all.py` as the single-entry point.
- `figures/` and `tables/` generated automatically from code.

5 Results

5.1 Data preparation outcomes

After excluding zero-ADR records, the analytic dataset contains 117,430 bookings. The withADR matrix has 7 features and the noADR matrix has 6 features. PCA on the withADR representation indicates that the first principal component explains approximately 54.3% of the variance and that four components are needed to reach around 90% explained variance.

5.2 Model comparison

Table 4: Main model comparison: quantitative results.

Method	**Setting**	**Sil.**	**CH**	**DB**	**Stability**
K-Means	$k = 2$, withADR	0.392	54,947	1.074	ARI 0.998 ± 0.002
K-Means	$k = 3$, withADR	0.452	63,837	0.857	ARI 0.989 ± 0.007
K-Means	$k = 3$, noADR	0.443	60,587	0.928	n/a
iK-Means	skipped / unresolved	n/a	n/a	n/a	n/a
Ward	$k = 3$, withADR, $n = 30{,}000$	0.453	15,928	0.832	subsample stable

Table 5: Main model comparison: interpretation.

Model / setting	Observation
K-Means, $k = 2$, withADR	Stable but too coarse: it merges the Premium Resort Long-Stay profile with the Standard Flexible City Guest profile.
K-Means, $k = 3$, withADR	Selected solution. It gives the best balance between internal validity, seed stability, and business interpretability.
K-Means, $k = 3$, noADR	Useful sensitivity check. The commitment axis remains visible without price, but the separation between Standard and Premium profiles weakens.
iK-Means	Not selected. In the current uploaded experiment log, iK-Means is marked as skipped/unresolved, so final numerical claims should only be included after the notebook and `experiments.csv` are updated.
Ward, $k = 3$, withADR	Conceptually distinct method that corroborates the same three-cluster structure on a reproducible subsample.

The selected solution is K-Means with $k = 3$ on the withADR representation. This choice is justified because it achieves the strongest balance between internal validity, stability, and interpretability. Although $k = 2$ is stable, it merges two managerially different segments. Values above $k = 3$ tend to fragment the same business profiles without producing clearer narratives.

5.3 Cluster profiles

Table 6: Selected K-Means cluster profiles for $k = 3$ withADR.

Cluster	Size	%	Main characteristics	Label
C2	14,513	12.4	Non-refundable bookings, short stays, advance booking horizon, high revenue certainty	Non-Refundable Advance Booker
C1	83,139	70.7	Flexible bookings, short stays, city-hotel dominant, broad booking horizons, relatively higher parking need	Standard Flexible City Guest
C0	19,778	16.8	Longer stays, higher estimated spend, more special requests, resort-hotel dominant	Premium Resort Long-Stay

5.4 Cluster interpretation

The **Non-Refundable Advance Booker** segment is defined by deposit commitment and a more advanced booking horizon. Although its average daily rate is not necessarily the highest, the segment provides high revenue certainty because cancellation flexibility is limited.

The **Standard Flexible City Guest** segment forms the majority of bookings. These records are typically flexible, short-stay bookings with no non-refundable commitment. This segment likely contains both business and urban leisure demand.

The **Premium Resort Long-Stay** segment is the smallest but most valuable profile. It is characterised by longer stays, higher estimated total spend, and more special requests. This segment is suitable for pre-arrival personalisation and upselling.

5.5 noADR comparison

The noADR representation tests whether clusters are merely price bands. Removing `total_spend_log` reduces the explicit price/value signal. The Non-Refundable segment remains structurally meaningful, indicating that deposit commitment is an independent axis of segmentation. However, separation between Standard and Premium bookings becomes weaker because price and total spend are important for distinguishing high-value long-stay behaviour.

5.6 PCA representation study: Extension Module E4

Table 7: PCA representation study for K-Means with $k = 3$.

Representation	Variance explained	Silhouette	CH	ARI vs original
Original 7D	1.000	0.452	63,837	1.000
PCA-2	0.773	**TODO:** fill	**TODO:** fill	**TODO:** fill
PCA-3	0.883	**TODO:** fill	**TODO:** fill	**TODO:** fill
PCA-4	0.972	**TODO:** fill	**TODO:** fill	**TODO:** fill

The main methodological caveat is that preserving variance is not equivalent to preserving cluster-separating directions. In this dataset, the deposit-commitment feature has relatively low marginal variance but strong segmentation meaning. PCA may therefore compress behaviourally important dimensions into later components. The original 7D representation is retained unless PCA produces comparable ARI and internal validity without reducing interpretability.

6 Discussion

6.1 Which result best answers the segmentation question?

K-Means with $k = 3$ on the withADR representation best answers the segmentation question. It provides three stable and interpretable booking profiles that align with actionable revenue-management decisions: commitment-based rate locking, flexible city-hotel demand management, and high-value resort-stay personalisation.

6.2 Influence of preprocessing

The feature weight on `is_non_refund` is one of the most consequential preprocessing choices. Without weighting, the binary commitment signal is dominated by continuous features. The meal-plan weight reduces the effect of an ordinal encoding that might otherwise exaggerate differences between meal categories. These weights make the Euclidean metric more aligned with the business segmentation question, but they also introduce subjective assumptions.

6.3 Stability and sensitivity

K-Means is highly stable across seeds for the selected $k = 3$ solution. Hierarchical Ward clustering provides independent support for the three-cluster structure. Sensitivity checks using scaler changes and the noADR representation indicate that the main qualitative conclusions are not driven solely by a single preprocessing decision, although the distinction between Standard and Premium segments depends partly on value-related features.

6.4 Decision-relevant recommendations

- **Non-Refundable Advance Booker:** offer early-bird packages and prepaid upgrades at booking time.
- **Standard Flexible City Guest:** use targeted flexibility-based promotions, parking-inclusive packages, and last-minute offers.

- **Premium Resort Long-Stay:** prioritise pre-arrival personalisation, room upgrades, spa/F&B offers, and long-stay service planning.

6.5 What should not be concluded

The clusters are not externally validated customer types. They are booking-level geometric segments, not permanent guest identities. A returning guest may appear in different clusters across different stays. Internal validity indices measure geometric compactness and separation, not business profitability or causal effects.

7 Limitations, Ethics, and Reproducibility

7.1 Limitations

- There are no ground-truth segment labels.
- The unit of analysis is a booking record, not a unique guest.
- The meal-plan encoding assumes an ordinal structure and equal spacing between meal levels.
- Lead-time quantile binning simplifies a highly skewed continuous variable.
- Manual feature weights encode subjective modelling assumptions.
- The dataset concerns two Portuguese hotels from 2015–2017, limiting generalisability.
- iK-Means results must be reconciled across the notebook, report, and experiment log before final submission.

7.2 Ethics and governance

The dataset is used according to the course release terms. Raw data are not committed to the repository. No direct personal identifiers are used. Country is excluded from clustering inputs and retained only, if needed, for post-hoc descriptive analysis.

7.3 Reproducibility checklist

The final repository should contain:

- `README.md` with setup and execution instructions.
- `environment.yml` or equivalent lockfile.
- `run_all.py` as a single-entry point.
- `experiments.csv` with method, parameters, seeds, metrics, diagnostics, and notes.
- `src/`, `figures/`, and `tables/` folders.
- A `Submission` tag or GitHub release.

7.4 Academic integrity disclosure

Generative-AI tools were consulted only for methodological discussion, syntax review, and report-structure support. All final analytical decisions, modelling choices, experimental results, and interpretations must be validated by the team.

8 Conclusion

This study segments 117,430 hotel booking records using leakage-safe pre-stay features. The selected K-Means solution with $k = 3$ on the withADR representation identifies three interpretable booking profiles: Non-Refundable Advance Booker, Standard Flexible City Guest, and Premium Resort Long-Stay. The results are supported by internal validation, seed stability, and comparison with Ward hierarchical clustering. The noADR comparison shows that booking commitment remains an important segmentation axis independently of price, while the value-inclusive representation better answers the revenue-management question. Future work should validate the segments against business outcomes, test alternative encodings, and evaluate generalisability across other hotels and time periods.

A Feature Table

Table 8: withADR feature table.

Feature	Type	Scaling	Weight
`total_spend_log`	Continuous	RobustScaler	$\times 1.0$
`total_nights`	Continuous	RobustScaler	$\times 1.0$
`is_non_refund`	Binary	Raw	$\times 3.0$
`total_of_special_requests`	Count	Raw	$\times 1.0$
`required_car_parking_spaces`	Binary/count	Raw	$\times 1.0$
`meal_encoded`	Ordinal	Raw	$\times 0.5$
`lead_time_rank`	Ordinal	Raw	$\times 1.0$

B K-Means Stability Summary

Table 9: K-Means stability across 10 seeds.

k	Mean ARI	Std ARI	Mean Sil.	Std Sil.	Notes
2	0.9975	0.0021	**TODO:** fill	**TODO:** fill	Stable but under-segmented
3	0.9894	0.0075	**TODO:** fill	**TODO:** fill	Selected solution
4	0.9942	0.0057	**TODO:** fill	**TODO:** fill	Additional fragmentation
5	0.9803	0.0138	**TODO:** fill	**TODO:** fill	Less interpretable

C Scaler Sensitivity

Table 10: Scaler sensitivity for numerical features, $k = 3$.

Scaler	Silhouette	CH	DB
RobustScaler	**TODO:** fill	**TODO:** fill	**TODO:** fill
StandardScaler	**TODO:** fill	**TODO:** fill	**TODO:** fill

References

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